

Executive summary. This project classifies public CFPB complaint narratives into eight financial product categories and uses locked decision-score rules to recommend either automatic routing or human review. It combines leakage-safe model evaluation, a tested human-in-the-loop policy, and a one-time 2025 out-of-time validation. The system is an internal prototype, not a deployed production service.

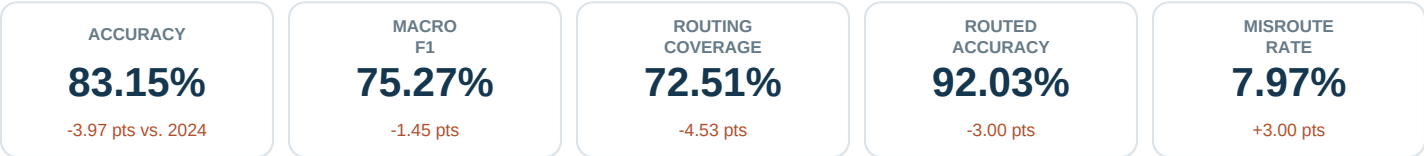
Business Problem

Complaint operations teams must route unstructured narratives to the correct product area. Fully manual triage can be slow and inconsistent, while uncontrolled automation can misroute complex or higher-risk cases. The objective is consistent initial recommendations with human review for weaker or ambiguous model signals.

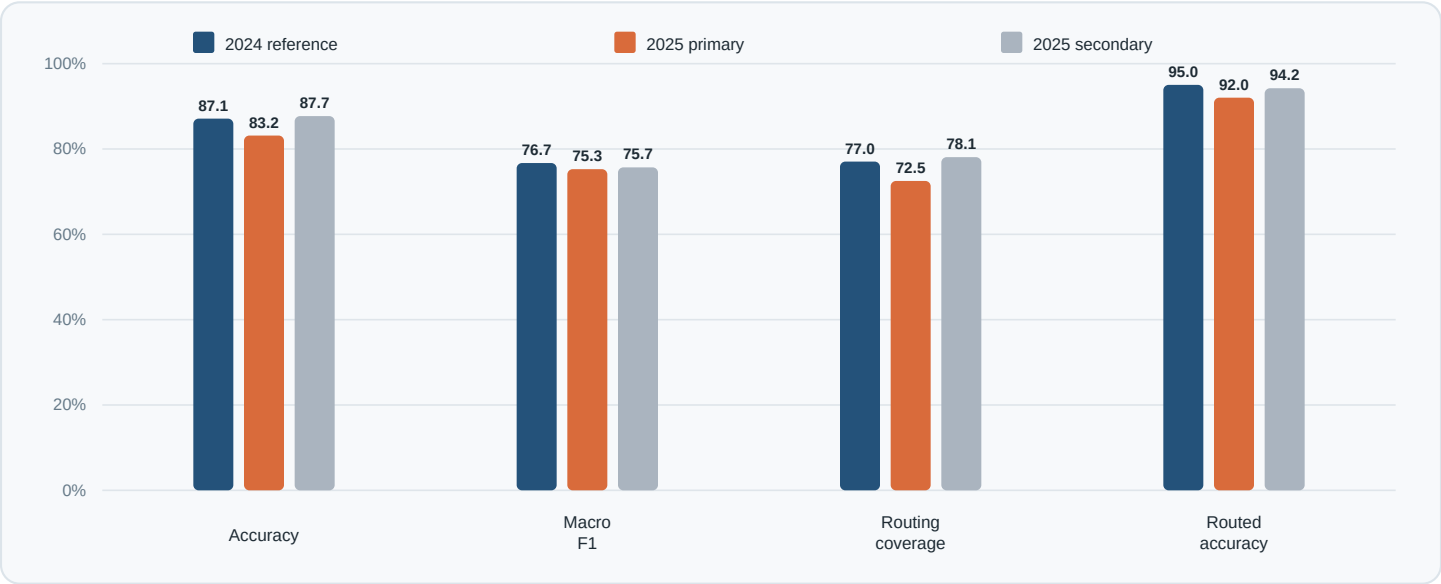
Solution Workflow



Headline 2025 Primary Results



Locked 2024 Reference vs. 2025 Cohorts



Headline finding: The locked workflow remained useful, but classification and routing performance weakened on the stricter 30,156-row leakage-resistant 2025 cohort. The more favorable 49,225-row secondary cohort is a sensitivity view because it retains repeated text and cross-year overlap.

Leakage-Safe Methodology

39.39%

superseded test rows overlapped training text

33,042

corrected modeling rows

6,609

locked final-test rows

0

normalized-text overlap groups

- 1

Audit. Detected duplicate-text contamination in the original row-level split.
- 2

Remediate. Excluded conflicting-label text groups and retained one representative from each repeated same-label group.
- 3

Split safely. Used group-aware development/test boundaries so normalized text could not cross partitions.
- 4

Select without test leakage. Compared three TF-IDF pipelines using development-only Macro F1.

Model and Routing Design

Model candidate	Dev CV Macro F1
Logistic Regression	0.7430
Multinomial Naive Bayes	0.3519
Linear SVM	0.7687

Locked routing rule	Value
Minimum top score	0.08
Minimum top-two margin	0.73

Both thresholds must pass. Scores and margins are model signals, not calibrated probabilities.

One-Time 2025 Out-of-Time Validation

A validation protocol was committed before 2025 access. The fitted model, preprocessing, 50,000-feature TF-IDF representation, eight classes, cohorts, metrics, and routing thresholds remained locked; no 2025 fitting, tuning, calibration, or selection occurred.

Cohort	Rows	Purpose
Primary leakage-resistant	30,156	Headline estimate; excludes 2024 text overlap, conflicts, and extra repetitions.
Secondary operational	49,225	Sensitivity view retaining repeated text and cross-year overlap.

Holdout status: Future changes require a new untouched validation period.

Key Business and Risk Findings

- Primary future performance weakened: accuracy, weighted F1, routing coverage, and routed accuracy declined.
- Human-review rate increased from 22.95% to 27.49%.
- Primary routed misroute rate increased from 4.97% to 7.97%.
- Category behavior was uneven: money-transfer F1 improved, while student-loan F1 declined.
- Aggregate metrics can hide higher category-level routing risk and small-sample instability.

Limitations and Next Steps

- Sampled public CFPB data may not represent a private operational queue.
- No probability calibration, fairness evaluation, production monitoring, or governance approval is claimed.
- Future work: category-specific error costs, stakeholder-approved review policies, calibration, DistilBERT comparison, secure deployment design, and a new validation period.

Technology and Project Artifacts

PythonPandasScikit-learnTF-IDFLinear SVMJupyter

MatplotlibGitHub Actions

Supporting artifacts include the leakage-safe modeling notebook, decision-score routing analysis, pre-holdout protocol, 2025 validation notebook, model card, detailed results report, and tested routing-rule module.

Explore the repository
github.com/rihua-tech/financial-complaint-auto-routing-nlp